

Resources ▼

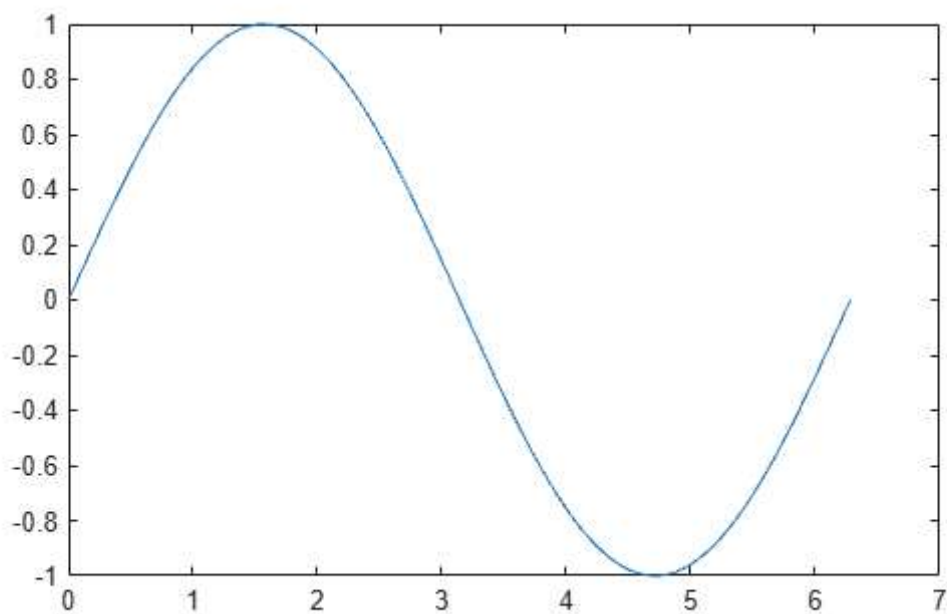
2-D and 3-D Plots

Line Plots

To create two-dimensional line plots, use the `plot` function. For example, plot the sine function over a linearly spaced vector of values from 0 to 2π :

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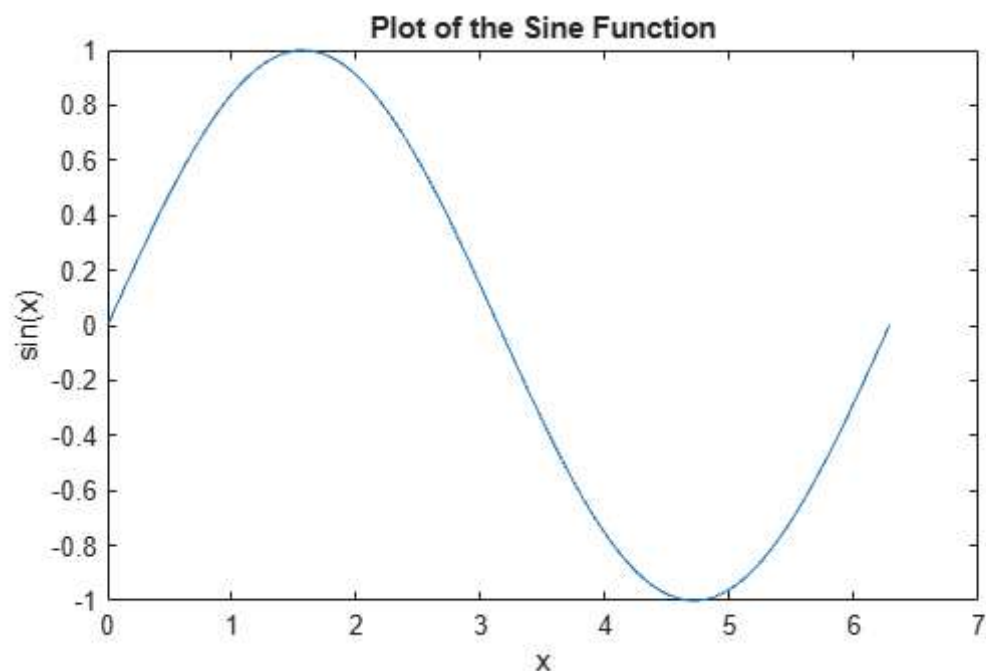
```
x = linspace(0,2*pi);  
y = sin(x);  
plot(x,y)
```

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You can label the axes and add a title.

```
xlabel("x")  
ylabel("sin(x)")  
title("Plot of the Sine Function")
```

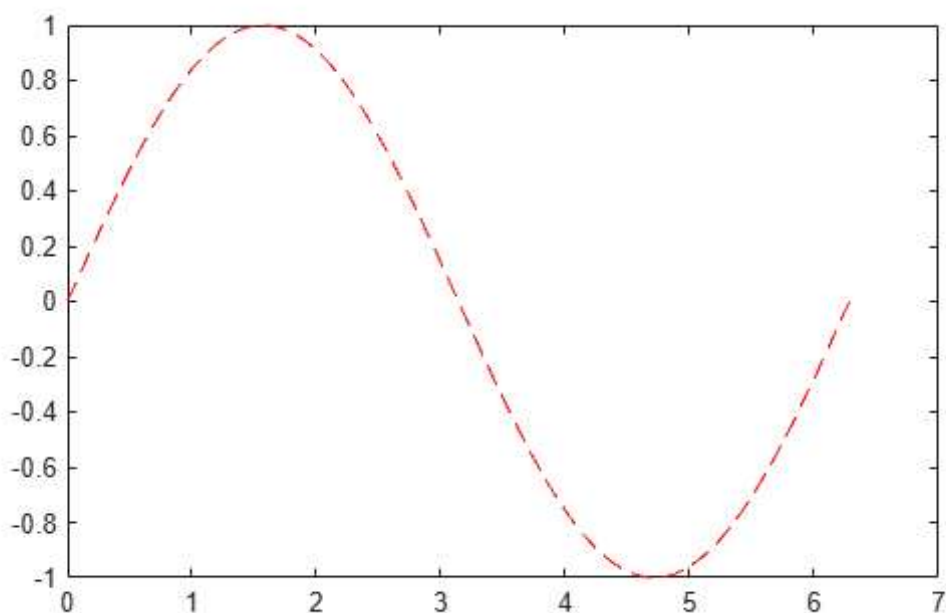
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By adding a third input argument to the `plot` function, you can plot the same variables using a red dashed line.

```
plot(x,y,"r--")
```

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"r--" is a *line specification*. Each specification can include characters for the line color, style, and marker. A marker is a symbol that appears at each plotted data point, such as a +, o, or *. For example, "g:*" requests a dotted green line with * markers.

Notice that the titles and labels that you defined for the first plot are no longer in the current figure window. By default, MATLAB® clears the figure each time you call a plotting function, resetting the axes and other elements to prepare the new plot.

To add plots to an existing figure, use `hold on`. Until you use `hold off` or close the window, all plots appear in the current figure window.

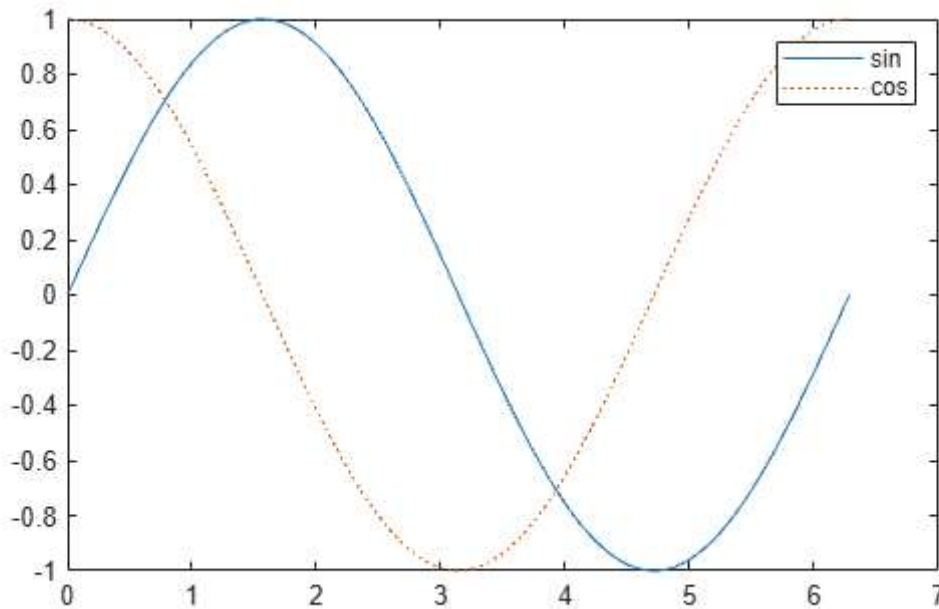
```
x = linspace(0,2*pi);
y = sin(x);
plot(x,y)
```

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```
hold on
```

```
y2 = cos(x);
plot(x,y2,":")
legend("sin","cos")
```

```
hold off
```



3-D Plots

Three-dimensional plots typically display a surface defined by a function in two variables, $z = f(x, y)$. For instance, calculate $z = xe^{-x^2-y^2}$ given row and column vectors x and y with 20 points each in the range $[-2,2]$.

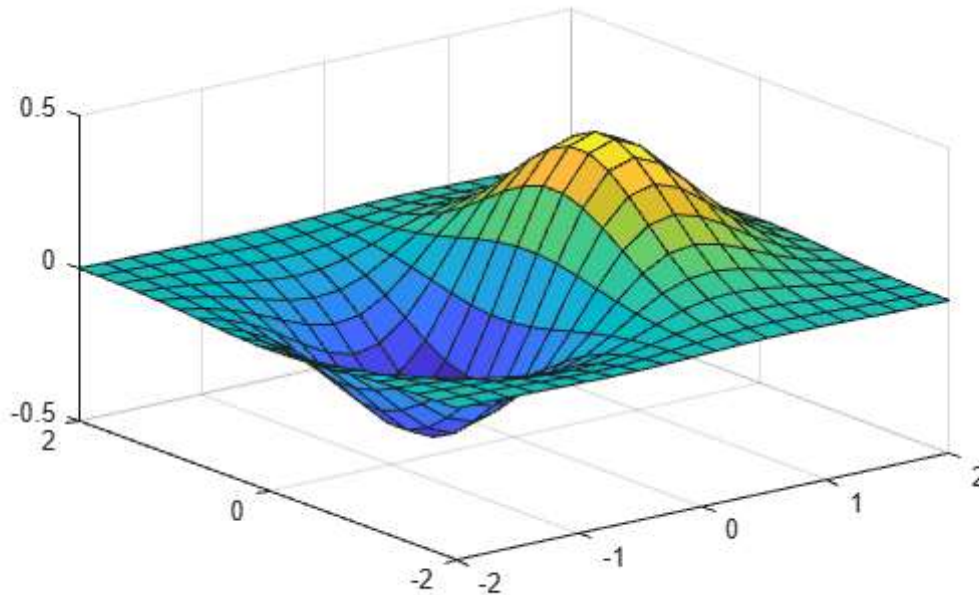
```
x = linspace(-2,2,20);
y = x';
z = x .* exp(-x.^2 - y.^2);
```

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Then, create a surface plot.

```
surf(x,y,z)
```

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Both the `surf` function and its companion `mesh` display surfaces in three dimensions. `surf` displays both the connecting lines and the faces of the surface in color. `mesh` produces wireframe surfaces that color only the connecting lines.

Multiple Plots

You can display multiple plots in different parts of the same window using either `tiledlayout` or `subplot`.

The `tiledlayout` function was introduced in R2019b and provides more control over labels and spacing than `subplot`. For example, create a 2-by-2 layout within a figure window. Then, call `nexttile` each time you want a plot to appear in the next region.

```
t = tiledlayout(2,2);
title(t,"Trigonometric Functions")
x = linspace(0,30);

nexttile
plot(x,sin(x))
title("Sine")

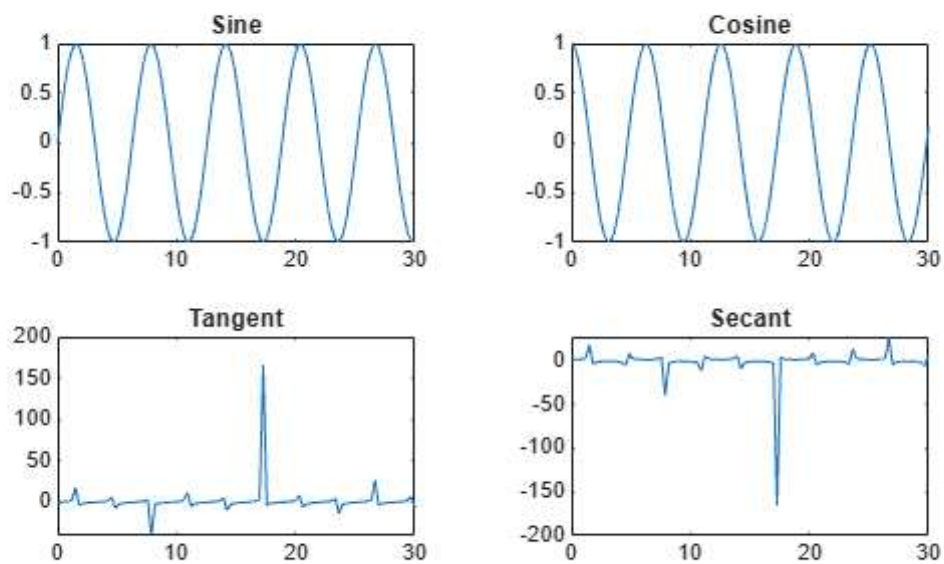
nexttile
plot(x,cos(x))
title("Cosine")

nexttile
plot(x,tan(x))
title("Tangent")

nexttile
plot(x,sec(x))
title("Secant")
```

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Trigonometric Functions



If you are using a release earlier than R2019b, see [subplot](#).

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